



# **J.D. COLLEGE OF ENGINEERING & MANAGEMENT**

# **PRESENTATION**

**ON**

**I/O PERFORMANCE IMPROVEMENT: SPOOLING,  
BUFFERING & CACHING**

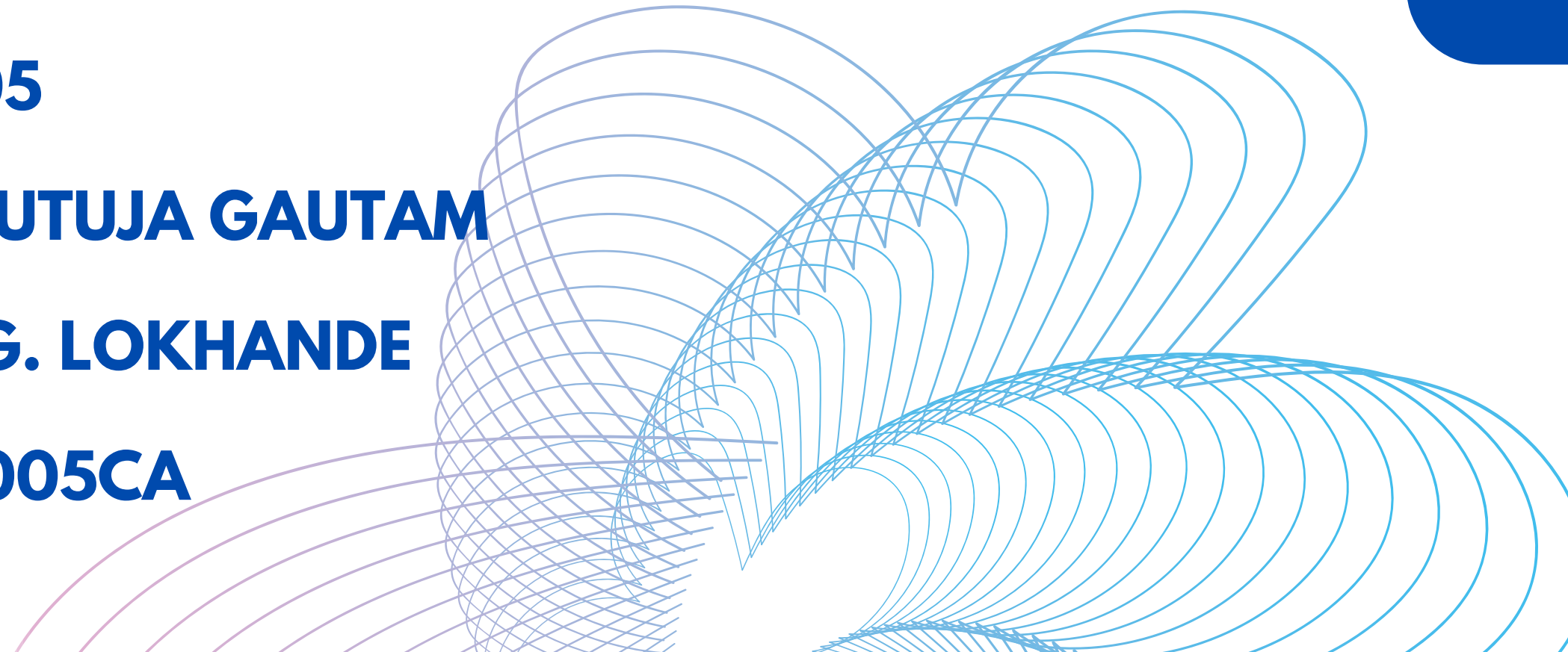
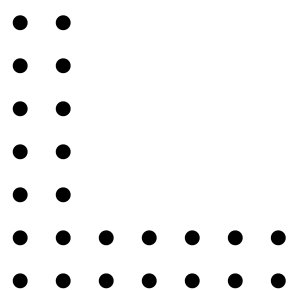
**SUBJECT : COMPUTER ARCHITECTURE AND ORGANIZATION**

**COURSE CODE : MC1A005**

**GUIDED BY : PROF. RUTUJA GAUTAM**

**PRESENTED BY : ADESH G. LOKHANDE**

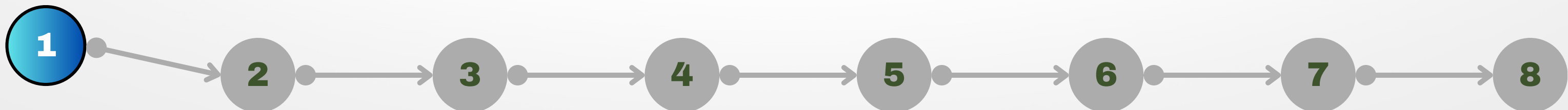
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# INTRODUCTION

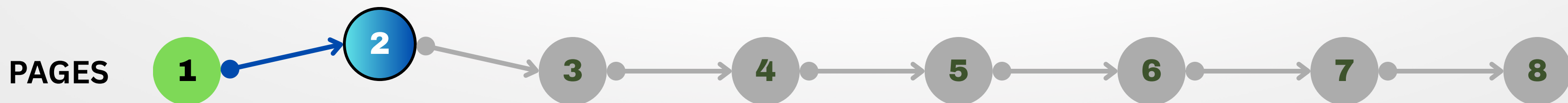
**Input and Output devices work at different speeds. A slow device can reduce overall system performance.**

**To solve this, Spooling, Buffering, and Caching techniques are used to improve I/O performance.**



# WHAT IS SPOOLING?

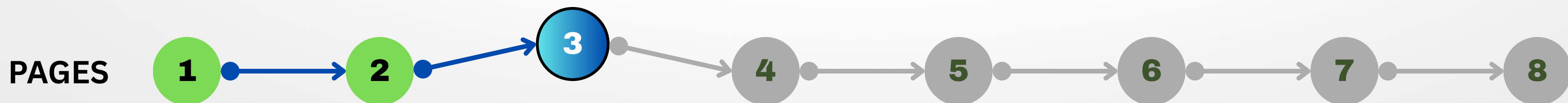
- Spooling stands for **Simultaneous Peripheral Operations Online**.
- It helps manage slow devices by storing work in a queue on the disk.
- The device performs tasks at its own speed while the CPU continues other work.



# EXAMPLE OF SPOOLING

- Printers are slower than computers.
- Print tasks are first stored in a queue on the disk.
- The printer prints them one by one without stopping the computer from working.

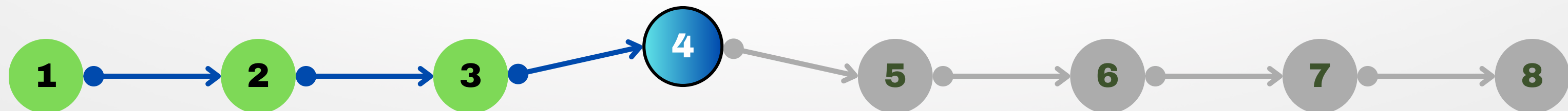
**Work is queued to avoid waiting for slow devices.**



# WHAT IS BUFFERING?

- Buffering temporarily stores data in a buffer in the RAM.
- It helps balance the speed difference between fast and slow devices.
- It keeps data flow smooth and prevents delays.

PAGES





# EXAMPLE OF BUFFERING

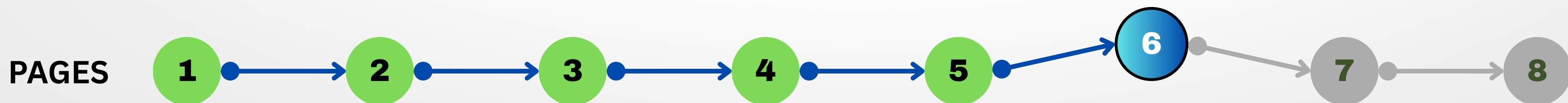
- When you play a YouTube video, some part is loaded first before playing.
- This stored data is called a buffer.
- It prevents pauses when data is being fetched.

**Data is held temporarily to avoid interruptions.**



# WHAT IS CACHING

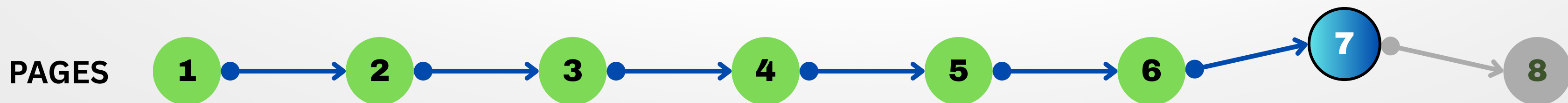
- **Caching stores frequently used data in cache memory, which is very close to the CPU.**
- **This reduces the need to fetch data repeatedly from slower memory.**
- **It increases the system performance.**



# EXAMPLE OF CACHING

- If you open a recently used file, it opens faster because it is stored in cache.
- The CPU can quickly access cached data.

**Frequently accessed data is stored in fast memory for quick access.**





# CONCLUSION

- 1** All three techniques improve I/O performance.
- 2** Spooling uses queue storage, Buffering holds data temporarily, and Caching stores frequently used data in fast memory.
- 3** These methods make the system faster and more efficient.





**THANK YOU**

**SO MUCH MA'AM TO GIVING ME  
THIS OPPORTUNITY**